

Spectrum Sense:

Autism Spectrum Disorder Risk Analysis



Presented by
Group-1 Sem-8 (SE)

CST-8316
Data Analysis and Management



MEMBERS

Kaung Myat Thu	TNT-1789
Shun Lae Thu	TNT-1804
Hnin Shwe Yi Wint	TNT-1853
Eaint Pyae Phyo	TNT-1886



Roles and Responsibilities

Kaung Myat Thu

Backend Developer

- Design database
- Database management
- Support backend logic
- Connect endpoints
- Testing
- Handle Errors

Shun Lae Thu

Frontend Developer

- Design project scope
- Support frontend logic
- Support UI/UX enhancement
- Connect endpoints
- Testing
- Handle Errors

Hnin Shwe Yi Wint

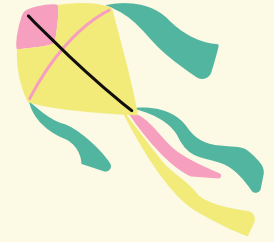
Frontend Developer

- Design structure and scope
- Support frontend logic
- Support UI/UX enhancement
- Connect endpoints
- Collect questionnaires
- Testing
- Handle Errors

Eaint Pyae Phyo

Model Trainer

- Design project scope
- Collect datasets
- Train ASD prediction models
- Implement API
- Testing
- Handle Errors

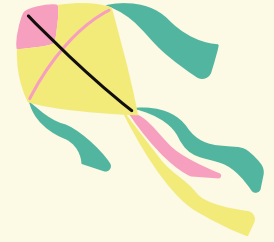


OBJECTIVES

- To design a **machine-learning-assisted** early ASD screening platform
- To ensure **age-appropriate** evaluation using **standardized tools**
- To analyze **behavioral** and **demographic data** to identify patterns associated with ASD risk.
- To provide **parent-friendly visualizations** that are easy to understand and interpret.
- To support **early awareness, intervention decisions** and check **developmental milestones**.
- To **reduce** ASD-related **misconceptions**, and **misinformation** through educational content.

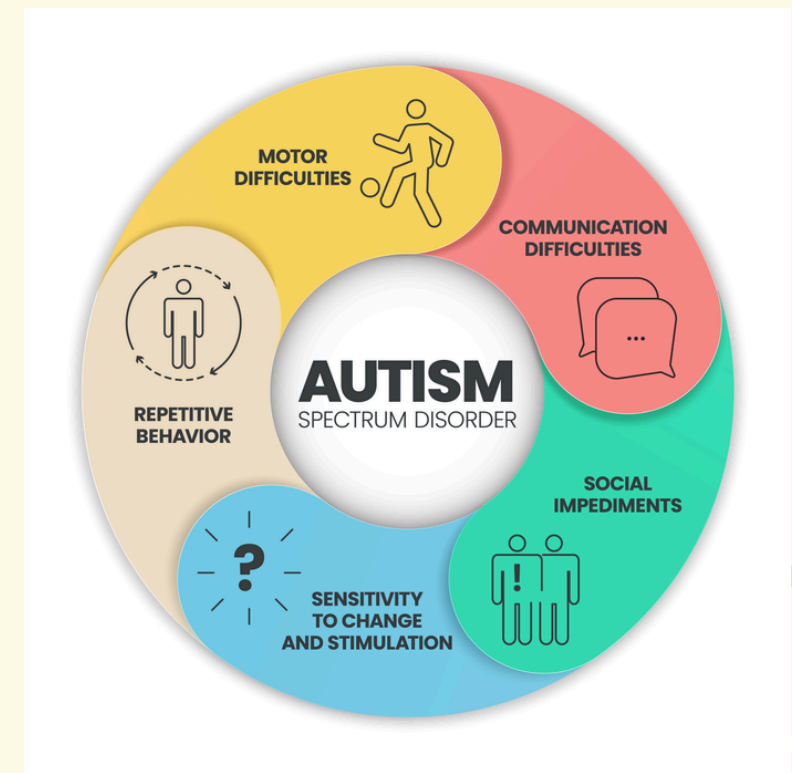
INTRODUCTION

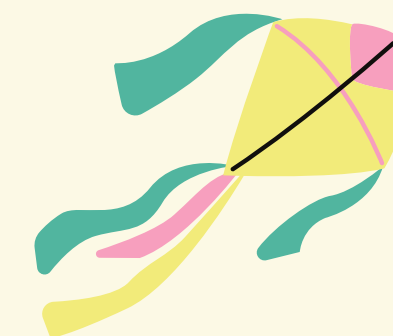
- This project proposes a **machine-learning-based ASD screening platform** designed to assist in **early identification of autism-related traits** in children.
- The system integrates **standardized screening tools** with **predictive analytics** to provide **age-appropriate assessments** and **automated risk evaluation**.
- The primary goal is to **enhance early detection accessibility while maintaining clinical awareness** that the system functions as **a screening support tool rather than a diagnostic replacement**.



What is Autism Spectrum Disorder (ASD)?

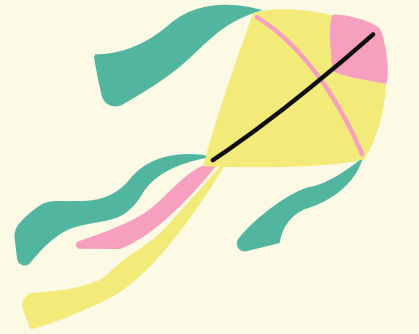
- Autism was **first described** by Leo Kanner in 1943.
- A complex **neurodevelopmental disorder** that begins in early childhood.
- Affects brain development related to **social interaction** and **communication**.
- ASD is a **spectrum disorder**, meaning:
 - **Symptoms** vary widely
 - **Severity differs** from person to person
 - Statistically, **boys** are approximately **4 times more** likely to be **diagnosed** than **girls**.





Autism Risk Factors

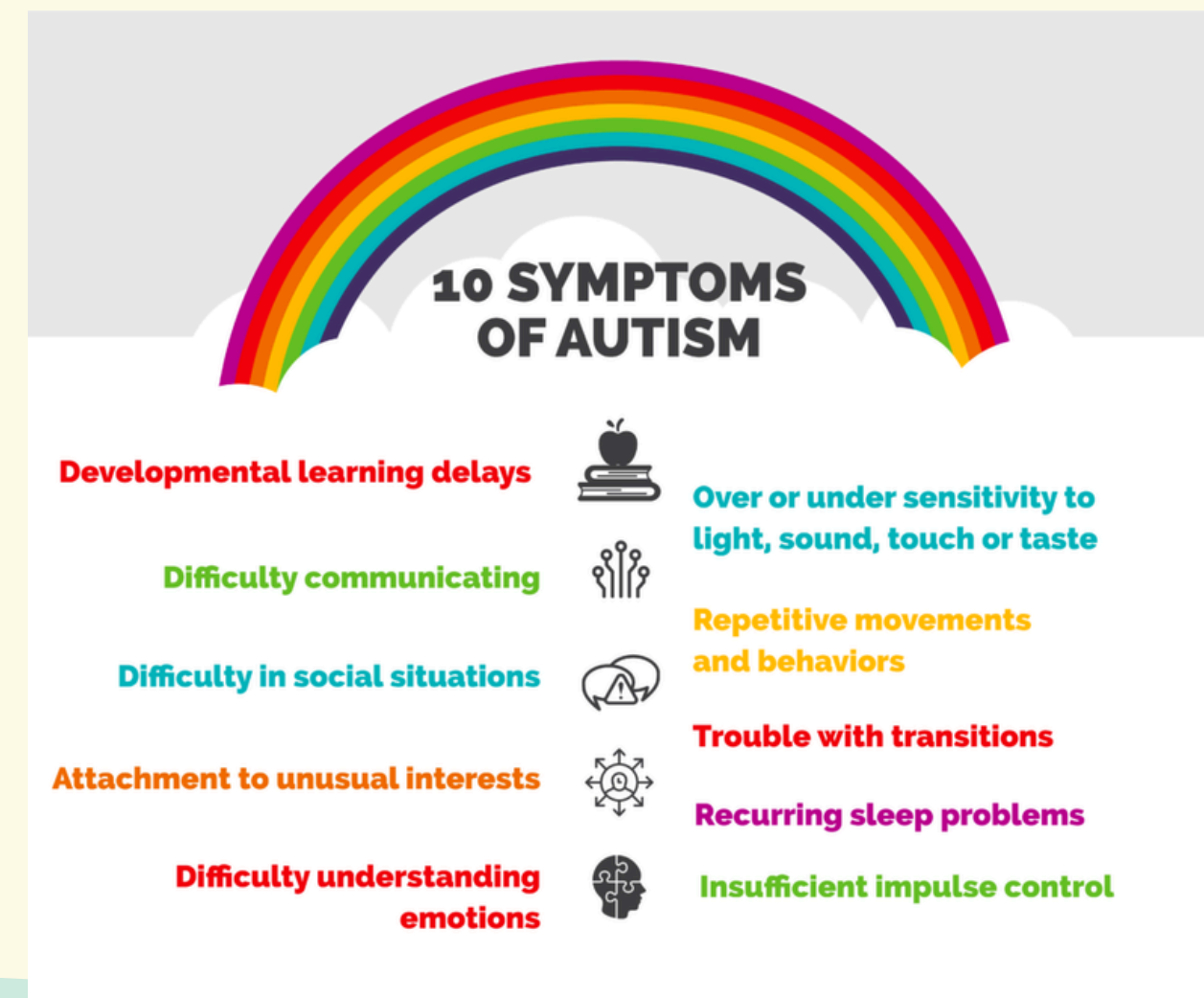


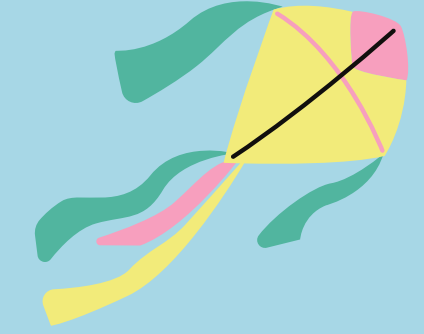


Autism Symptoms

The 10 most common symptoms of autism are:

1. Developmental learning delays
2. Over or under sensitivity to lights, sounds, touch, or tastes
3. Difficulty communicating
4. Repetitive movements and behaviors
5. Difficulty in social situations
6. Trouble with transitions
7. Attachment to unusual interests
8. Difficulty understanding emotions
9. Recurring sleep problems
10. Insufficient impulse control



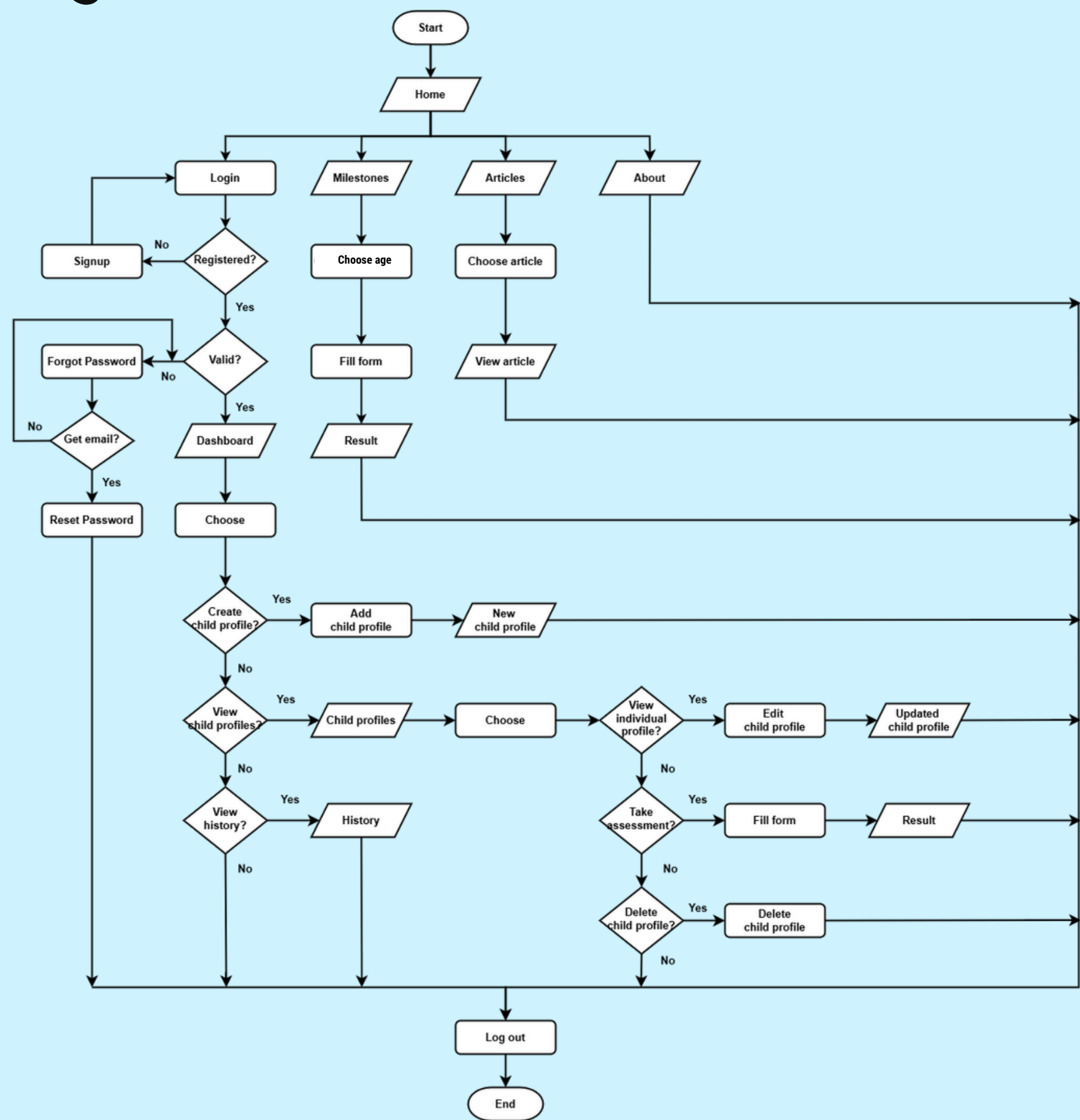


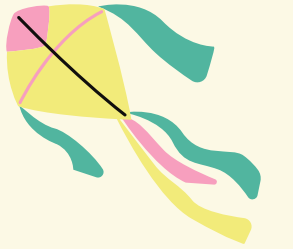
Target Audience

- Designed for parents, guardians, and caregivers, and healthcare professionals for better understanding a child's developmental patterns.
- Useful for identifying concerns in social communication
- Assists in observing behavioral patterns
- Aids in monitoring developmental milestones

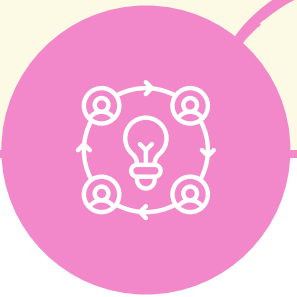


System Flow Diagram



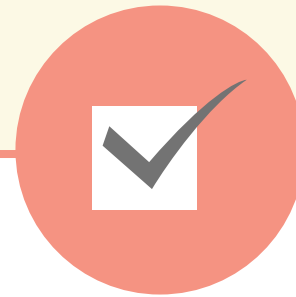


Key Features



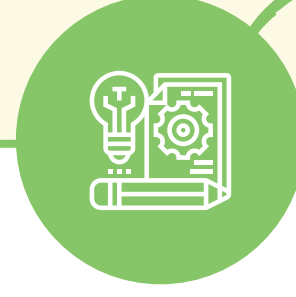
ASD Educational Content

Access **reliable articles and information** related to Autism Spectrum Disorder



Milestones Check

Parent-completed check lists based on age-specified developmental milestones



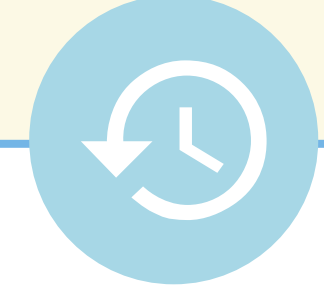
Screening Questionnaires

Parent-completed questionnaires based on standardized ASD screening datasets



Child Profile Management

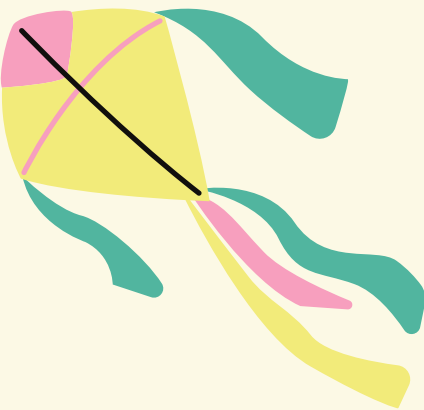
Create and manage **child profiles** (age, gender, region)



Analysis History

View previous screening **results** summaries over time either by overall or individual





M-CHAT (Modified Checklist for Autism in Toddlers)

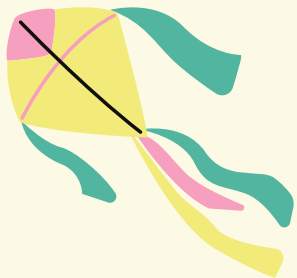
- developed by psychologists in 2000s to get the basic that this child has ASD or not
- Designed for early autism screening in toddlers (18-48 months)
- Parent-reported Yes/No questionnaire
- Screening tool (not diagnostic)

M-CHAT (Modified Checklist for Autism in Toddlers)

Please fill out the following about how your child **usually** is. Please try to answer every question. If the behavior is rare (e.g., you've seen it once or twice), please answer as if the child does not do it.

1.	Does your child enjoy being swung, bounced on your knee, etc.?	Yes	No
2.	Does your child take an interest in other children?	Yes	No
3.	Does your child like climbing on things, such as up stairs?	Yes	No
4.	Does your child enjoy playing peek-a-boo/hide-and-seek?	Yes	No
5.	Does your child ever pretend, for example, to talk on the phone or take care of dolls, or pretend other things?	Yes	No
6.	Does your child ever use his/her index finger to point, to ask for something?	Yes	No
7.	Does your child ever use his/her index finger to point, to indicate interest in something?	Yes	No
8.	Can your child play properly with small toys (e.g. cars or bricks) without just mouthing, fiddling, or dropping them?	Yes	No
9.	Does your child ever bring objects over to you (parent) to show you something?	Yes	No
10.	Does your child look you in the eye for more than a second or two?	Yes	No
11.	Does your child ever seem oversensitive to noise? (e.g., plugging ears)	Yes	No
12.	Does your child smile in response to your face or your smile?	Yes	No
13.	Does your child imitate you? (e.g., you make a face-will your child imitate it?)	Yes	No
14.	Does your child respond to his/her name when you call?	Yes	No
15.	If you point at a toy across the room, does your child look at it?	Yes	No
16.	Does your child walk?	Yes	No
17.	Does your child look at things you are looking at?	Yes	No
18.	Does your child make unusual finger movements near his/her face?	Yes	No
19.	Does your child try to attract your attention to his/her own activity?	Yes	No
20.	Have you ever wondered if your child is deaf?	Yes	No
21.	Does your child understand what people say?	Yes	No
22.	Does your child sometimes stare at nothing or wander with no purpose?	Yes	No
23.	Does your child look at your face to check your reaction when faced with something unfamiliar?	Yes	No





AQ-Child (Autism Spectrum Quotient – Child Version)

- Developed by **NovoPsych**, a digital health platform used by practitioners, private practices, and government & non-government health services.
- Designed for early autism screening in children (4-11 years)
- Parent-rated questionnaire
- Measures autism-related traits (Social, Attention Switching, Attention to details, Communication, Imagination)
- 4-point Likert scale (Definitely Agree, Slightly Agree, Slightly Disagree, Definitely Disagree)

Autism Spectrum Quotient Children’s Version (AQ-Child)

Instructions:

Please choose the response that best describes how strongly each statement applies to the child.

	Definitely Agree	Slightly Agree	Slightly Disagree	Definitely Disagree
1 They prefer to do things with others rather than on their own	0	1	2	3
2 They prefer to do things the same way over and over again	3	2	1	0
3 If they try to imagine something, they find it very easy to create a picture in their mind	0	1	2	3
4 They frequently get so strongly absorbed in one thing that they lose sight of other things	3	2	1	0
5 They often notice small sounds when others do not	3	2	1	0
6 They usually notice house numbers or similar strings of information	3	2	1	0
7 They have difficulty understanding rules for polite behaviour	3	2	1	0
8 When they are reading a story, they can easily imagine what the characters might look like	0	1	2	3
9 They are fascinated by dates	3	2	1	0
10 In a social group, they can easily keep track of several different people’s conversations	0	1	2	3
11 They find social situations easy	0	1	2	3
12 They tend to notice details that others do not	3	2	1	0
13 They would rather go to a library than a birthday party	3	2	1	0
14 They find making up stories easy	0	1	2	3
15 They are drawn more strongly to people than to things	0	1	2	3
16 They tend to have very strong interests, which they get upset about if they cannot pursue	3	2	1	0
17 They enjoy social chit-chat	0	1	2	3



Risk Classification

After submission, the system:

- Processes responses using a trained model
- Classifies results into:
 -  Low Risk >>> Below 30%
 -  Medium Risk >>> Between 30% and 70%
 -  High Risk >>> Above 70%
- ASD Risk Probability (percentage)

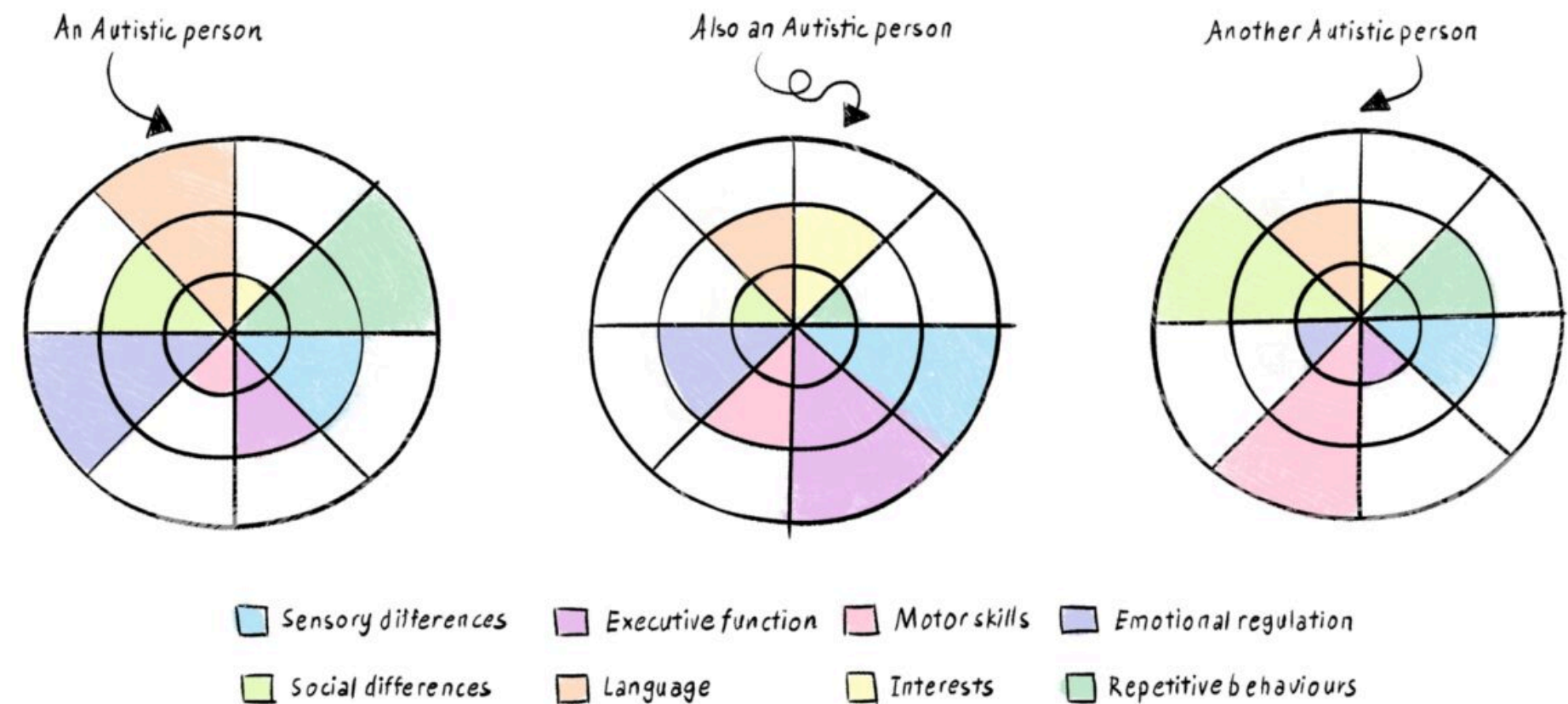
Visual Representation

 Spider Wheel (Radar Chart)

- Displays trait distribution across behavioral domains
- Helps visualize traits intensity

 Traits Analysis (Ring Chart)

- Shows risk percentage comparison
- Provides intuitive visual interpretation



Each color = A trait

Every autistic person has all the colors but **different intensities**

DATASETS

Dataset 1: M-CHAT Dataset

Purpose: Early ASD risk screening (binary classification)

Total Sample (N): 8036 rows

Features:

- 23 behavioral questionnaire responses (Q1–Q23)
- Demographic attributes (Age, Gender, Jaundice, Family ASD History)

Target Variable:

- ASD Risk = 1 (Positive)
- ASD Risk = 0 (Negative)

Dataset 2: AQ-Child Dataset

Purpose: ASD risk prediction (binary classification)

Total Sample (N): 10,088 rows

Features:

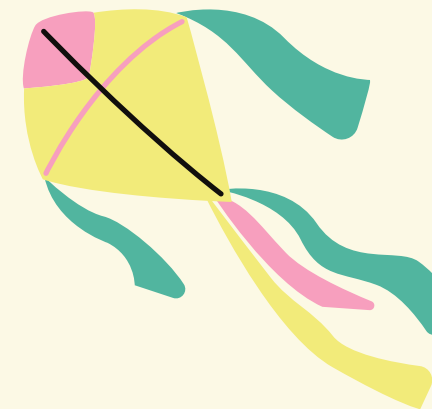
- 30 behavioral questionnaire responses (Q1–Q30)
- Demographic attributes (Age, Gender, Jaundice, Family ASD History)

Target Variable:

- ASD = 1 (Positive)
- ASD = 0 (Negative)



Algorithm Overview – ASD Risk Prediction System



1 Data Input

- Questionnaire responses (AQ / M-CHAT)
- Demographic features (Age, Gender, Family History, etc.)

2 Preprocessing

- Text standardization
- Categorical encoding
- Validation of score ranges
- Removal of duplicates & invalid entries

3 Feature Engineering

- Subscale mean scores
- Total risk scores
- High-score counts
- Interaction features (e.g., Social × Communication)

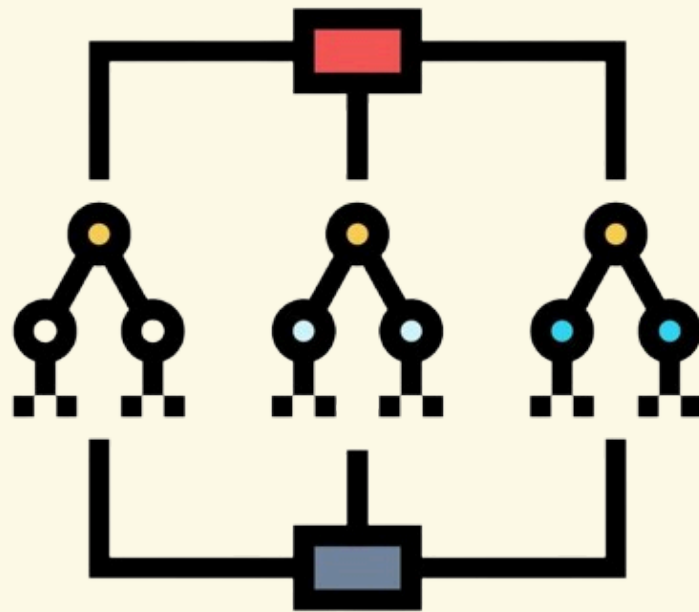
4 Model Prediction

- Random Forest classifier
- Probability estimation
- Threshold optimization
- Risk categorization (Low / Medium / High)

Random Forest Algorithm

Why Random Forest?

- Handles tabular questionnaire data well
- Captures nonlinear relationships
- Robust to noise
- Provides feature importance
- Works well with moderate dataset sizes



How It Works

1. Creates multiple decision trees (300–400 trees)
2. Each tree is trained on a random subset of data
3. Each tree gives a class prediction
4. Final probability = proportion of trees voting ASD

$$P(\text{ASD}) = \frac{\text{Number of Trees Predicting ASD}}{\text{Total Number of Trees}}$$

MODEL OPTIMIZATION STRATEGY

Hyperparameter Tuning

- Grid Search with 5-fold CV
- F2-Score optimization
- Tested:
 - class_weigh
 - max_depth
 - min_samples_split
 - min_samples_leaf
 - max_features
 - n_estimators (300–400)

Why F2-Score?

- Recall weighted higher than precision
- Prioritizes catching ASD cases
- Suitable for screening systems

M-CHAT Model

```
✓ Training complete!  
Best CV F2-Score: 0.7539  
  
Best Hyperparameters:  
class_weight: balanced  
max_depth: 25  
max_features: sqrt  
min_samples_leaf: 4  
min_samples_split: 10  
n_estimators: 300
```

AQ Model

```
✓ Training complete!  
Best CV F2-Score: 0.8179  
  
Best Hyperparameters:  
class_weight: balanced  
max_depth: 35  
max_features: log2  
min_samples_leaf: 4  
min_samples_split: 10  
n_estimators: 400
```



Threshold Optimization for Clinical Sensitivity

M-CHAT Model

Threshold range: 0.30 – 0.50

Selected threshold: 0.30

Reason: Maximized F2-score

AQ Model

Testing different classification thresholds:					
Threshold	Accuracy	Precision	Recall	F1	F2
0.30	0.7858	0.7131	0.8160	0.7611	0.7931
0.35	0.8108	0.7709	0.7791	0.7750	0.7775
0.40	0.8204	0.8019	0.7577	0.7792	0.7661
0.45	0.8242	0.8281	0.7316	0.7769	0.7491
0.50	0.8326	0.8614	0.7147	0.7812	0.7399

Testing different classification thresholds:					
Threshold	Accuracy	Precision	Recall	F1	F2
0.30	0.7976	0.7281	0.9156	0.8112	0.8708
0.35	0.8153	0.7595	0.8942	0.8214	0.8636
0.40	0.8239	0.7845	0.8675	0.8239	0.8496
0.45	0.8311	0.8111	0.8397	0.8252	0.8339
0.50	0.8265	0.8278	0.8013	0.8143	0.8065

Final M-CHAT Performance

- Recall: 81.6%
- Precision: 71.3%
- ROC-AUC: 0.8731

	Predicted	
	No ASD	ASD
Actual No ASD	693	214
Actual ASD	120	532

*misses 18.4%

Final AQ Performance

- Recall: 91.6%
- Precision: 72.8%
- ROC-AUC: 0.9128

	Predicted	
	No ASD	ASD
Actual No ASD	715	320
Actual ASD	79	857

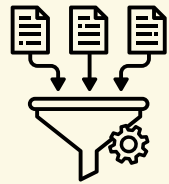
*misses 8.4%

TECHNOLOGY STACK



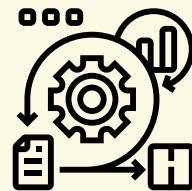
Data Storage

MongoDb (MongoDB Atlas)



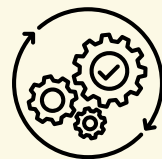
Data processing and analysis

Python (Pandas, Scikit-learn, NumPy)



Analytics & Algorithms

Random forest: Classification



Backend

NodeJs (Express)



Frontend

Typescript, Next.js, CSS

LIMITATIONS

Limited Training Data

- Many ASD datasets are small and not representative of all populations.

Not a Clinical Diagnosis Tool

- Final diagnosis must be done by qualified professionals (e.g., child psychologists, neurologists).

Behavioural Variability

- High-functioning individuals may be harder to detect.
- Symptoms change with age.

Limited Input Features

- May be subjective
- May contain reporting bias
- May not capture neurological complexity

Missclassification

- False positives → unnecessary stress for families
- False negatives → delayed intervention

FUTURE ENHANCEMENT

Integrate confirmed diagnosis feedback

- Collect ASD / Non-ASD results from users who receive hospital diagnosis after screening
- Use this verified data to retrain and improve model accuracy.

Expand authenticated dataset

- Gather larger, high-quality patient data from reliable and ethical sources to enhance model reliability and generalization.

Implement advanced models

- Compare multiple machine learning algorithms and select the best-performing model based on evaluation metrics.

Collaborate with medical professionals

- Work with pediatricians and developmental specialists to validate screening logic and improve system credibility.
- Analyze and refine questionnaire attributes to optimize prediction performance.

Parent-Centered Support Features

- Behavioral management guidance
- Mental health support resources for parents

CONCLUSION

- In conclusion, this project shows how technology can support early awareness of Autism Spectrum Disorder.
- By assisting in early screening, our system has the potential to promote faster intervention, better support, and improved outcomes for children and families.



REFERENCE

- The Autism Spectrum Quotient: Children's Version (AQ-Child) Article
- <https://www.altogetherautism.org.nz/what-is-autism/>
- <https://www.autismspeaks.org/autism-screening>
- <https://www.appliedbehavioranalysisprograms.com/lists/5-symptoms-of-autism/>

THANK YOU!

